

King's College London

This paper is part of an examination of the College counting towards the award of a degree. Examinations are governed by the College Regulations under the authority of the Academic Board.

Examination Period January 2024 (Period 1)
Module Code 6CCS3HCI
Module Title Human-Computer Interaction

Format of Examination Written questions
Start time 11am BST (GMT +1)
Time Allowed Two hours
Instructions You are permitted to access any materials you wish, but this is not mandated and is not expected. You may use a calculator if you find this helpful.
Rubric ANSWER ALL QUESTIONS

The rubric for this paper must be followed and extra answers should not be submitted. For answers that are handwritten, write with blue/black ink on light coloured paper. Include the Module code, question number and student number on every page to be submitted. For answers that are typed, use the template provided.

Submission Deadline
Submission Process Work must be submitted to the **level 6** Informatics Assessments KEATS page.

Your work must be submitted as a PDF file. If you have prepared some answers on computer, and some on paper (which have then been digitised), you may upload at most two PDF files – one for computer-prepared answers, one for digitised answers. Do not duplicate answers across the two PDFs – if you do this, the computer-prepared answer will be taken. You should check that your work displays correctly after it has been uploaded.

ACADEMIC HONESTY AND INTEGRITY: Students at King's are part of an academic community that values trust, fairness and respect and actively encourages students to act with honesty and integrity. It is a College policy that students take responsibility for their work and comply with the university's standards and requirements.

By submitting this assignment, I confirm that this work is entirely my own, or is the work of an assigned or permitted group, of which I am a member, with exception to any content where the works of others have been acknowledged with appropriate referencing. I also confirm that I have read and understood the College's Academic Honesty & Integrity Policy:

<https://www.kcl.ac.uk/governancezone/assessment/academic-honesty-integrity>

Misconduct regulations remain in place during this period and students can familiarise themselves with the procedures on the College website at

<https://www.kcl.ac.uk/campuslife/acservices/academic-regulations/assets-20-21/g27.pdf>

1. Foundational Concepts

- a. In a few sentences, describe some of the affordances and signifiers in Figure 1.

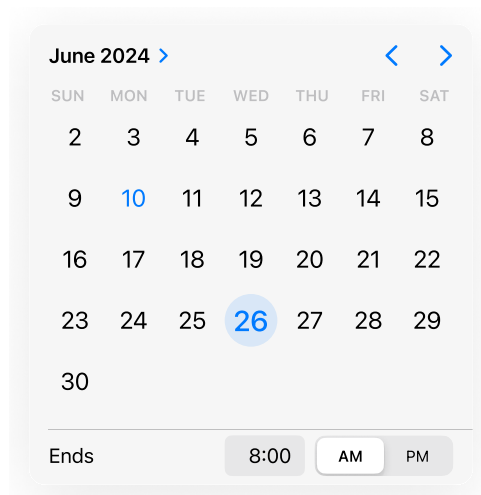


Figure 1: iOS Calendar Picker

[6 marks]

- b. Considering Figure 1 again, describe three of Nielsen's heuristics that it employs and why.

Important: there are many different correct answers here. We want to understand how you recall and apply the terminology.

[6 marks]

- c.** What is the difference between a conceptual and a mental model? Explain it through an example.

[6 marks]

- d.** What is the Hawthorne effect, and how might a user researcher mitigate against it? You can provide an example if it helps you explain.

[7 marks]

2. Empathise

- a. You have been hired by WalkMe, a social dog walking platform for dog owners wanting to meet up to walk their pets. You aim to improve the user experience for dog owners by better understanding their needs.

You are to interview 15 future users from your population. Assume you have already ethics, consent and have introduced the interviewee to the topic.

- i. Briefly describe your recruitment process and target demographic.
- ii. Draft an interview guide you could use, including 8-10 key questions you would ask.
- iii. Outline the key areas these questions will be covering – i.e., what aspects of the users' experience are you trying to learn more about and why.

[17 marks]

- b. Sketch an empathy map diagram and fill it out with 3 observations for each quadrant that you could imagine emerging from the interviews. Do not worry about what exactly your imagined observations say; we are trying to see how you are able to think about the aims of the empathy map process, rather than how well you imagine what 'real' users would say.

[8 marks]

3. Prototype

You are working for a company that makes high-end vacuum cleaners. The results of your empathise suggest that turning cleaning into a game makes the chore more exciting. Your ideate phase resulted in the 'HoloCleanse' feature, where the users use smart glasses while vacuuming and cleaning. In this way the chore is turned into a virtual reality game, where users pick up (virtual) coins on the floor with their vacuum.

- a. For your prototype to test if users like the idea of turning hoovering into a game, your manager has asked you to create a high-resolution prototype. Why might this be a bad idea?

[6 marks]

- b. Describe a Wizard of Oz prototype to test the minimum viable product. You may show simple sketches if you like.

[7 marks]

- c. Explain the difference between vertical and horizontal prototyping, choose the one you think is most suitable and state your reasoning.

[7 marks]

4. Evaluate

Your bank has developed a chatbot to help its customers troubleshoot issues. They want to know if their new prototype works better than their previous 'frequently asked questions' (FAQ) website.

a. Outline what would be useful evaluation metrics for your system:

- i. Identify 3 qualitative or quantitative indicators that you could use to show your prototype is useful compared to the previous service.
- ii. Explain why you selected these indicators, and how they might be measured.

[15 marks]

b. Outline two evaluation approaches you could use to answer the question in the context of this design problem. One should be more 'in the wild', and the other should be more controlled.

- i. Describe the rationale for each choice.
- ii. What questions would **not** be answered by each approach?

[15 marks]